Pixels, Frames & Transformers

Transfer Learning and Image Embeddings · Module 06

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Don’t Start from Scratch

A network trained on millions of images has already learned edges, textures, parts and objects. **Transfer learning** reuses that knowledge, just as BERT is reused for text.

1. Off the shelf
Use the model’s own labels: an ImageNet model names 1,000 categories, from *tabby cat* to *space shuttle*.

2. As an embedding machine
Remove the last layer, and the model turns every image into a vector. Train your own classifier on top, with your categories.

ImageNet: about 1.3 million training photos in 1,000 categories, the benchmark behind a decade of computer vision. Its categories were chosen by computer scientists: 120 of them are dog breeds, and none is *protest*, *politician* or *crowd*.

Cut Off the Last Layer: Image Embeddings



ResNet-50: a CNN trained *with labels* on ImageNet. 2,048 numbers per image.

DINOv2 (Oquab et al., 2024): a vision transformer trained *without labels* on 142 million images. 384 numbers per image (small version).

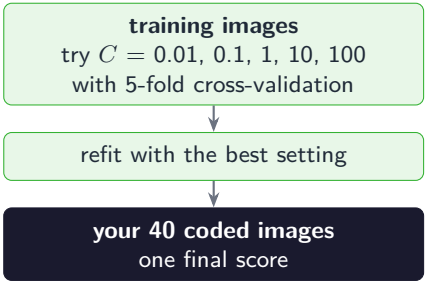
Images that look alike, to the model, get vectors that point in similar directions. With vectors of length 1, their dot product is the **cosine similarity**.

Three Ways to Use a Pre-trained Model

	What is trained	Cost	With few labels
Linear probe	a logistic regression on the frozen embeddings	seconds, no GPU once embedded	the safe default
Fine-tune the last block	the last block plus a new output layer	minutes on a GPU; more settings	sometimes better, often not
Fine-tune everything	every weight, at a low learning rate	the most compute and care	overfits easily

Start with a linear probe on good embeddings. It is the image version of a classifier on sentence embeddings, and usually far better than a CNN trained from scratch on the same few hundred labels. Fine-tune only when you have more labels and the probe has clearly hit a ceiling.

Choose Settings Without Touching the Test Set



Picking a setting by its score on the test set turns the test set into training data: you pick whatever happened to fit those 40 images, and the reported score is too optimistic.

Your scoreboard (`my_results.csv`) collects every method scored on the same 40 images: handcrafted features, a CNN from scratch, linear probes, fine-tuning, and later CLIP and vision-language models.

How Many Labels Do You Need?



schematic: mean and range over repeated random draws

A **learning curve**: train on random subsets of increasing size, several times per size, and score each on your codes.

- Good embeddings often get close to their plateau with a few dozen labels
- The band is wide at the small end: with 20 labels, *which* 20 you labeled matters a lot
- With 40 test images, every curve is noisy too

Use learning curves to decide whether more coding is worth it.

Search by Example, Find Near Duplicates

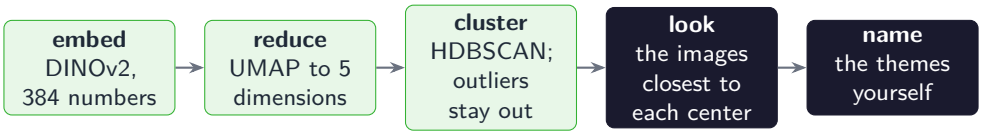
Search by example
Compute the cosine similarity between one image and all others, and look at the closest ones. No labels needed: a quick way to find more examples of a rare kind of image for coding.

Near duplicates
Pairs with very high similarity. Embeddings are trained to recognize the same scene under different crops, so they catch copies that perceptual hashes (module 02) miss.

copy	perceptual hash	embedding
resized, recompressed	✓	✓
brightened	✓	✓
cropped	only lightly	✓
mirrored	✗	✓
another photo, same moment	✗	close

Choose a threshold by **looking at pairs around it**, and decide what counts as a duplicate for your question.

Visual Themes: Embed, Reduce, Cluster, Look



Results depend on settings, as with topic models. Rerun with other cluster sizes and seeds; compare runs with the **adjusted Rand index**. Trust the themes that survive.

A cluster is not a frame. Clusters group what looks alike to the model: product type, brand, background. Frames come from theory (Torres, 2024). Use clusters to discover, a codebook to measure.

In the Notebook

- ImageNet labels off the shelf, for protest photos and packaging
- Embeddings from ResNet-50 and DINOv2
- Linear probes, a tuned probe and a fine-tuned ResNet-50, on your scoreboard
- A learning curve: how many labels do you need?
- Search by example and near duplicates
- Visual themes in cereal packaging, and how stable they are

part2_vectors/06_transfer_learning_and_embeddings/notebooks/
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